

16.8.

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{30} + \frac{1}{v} = \frac{1}{-10}$$

$$\text{解得 } v = -80\text{cm}$$

$$2x = 48\text{cm}$$

$$x = 24\text{cm}$$

16.13

$$(1) v = f_1 = 100\text{cm}$$

$$h' = f_1 \cdot \theta = 0.80\text{cm}$$

会聚透镜成在透镜右侧100cm的焦平面上

$$v = f_2 = -100\text{cm}$$

$$h' = |f_2| \cdot \theta = 0.80\text{cm}$$

发散透镜成在左侧100cm的焦平面上

$$(2) \frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

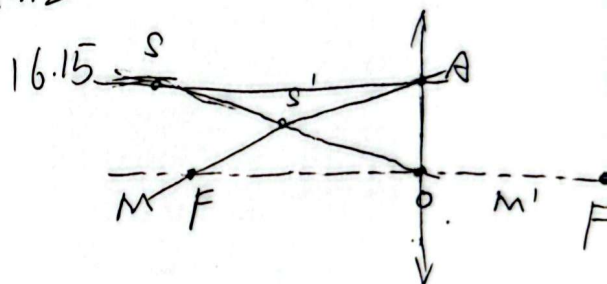
$$\frac{1}{10} + \frac{1}{v} = \frac{1}{10} \Rightarrow v = \infty$$

成像在无穷远处

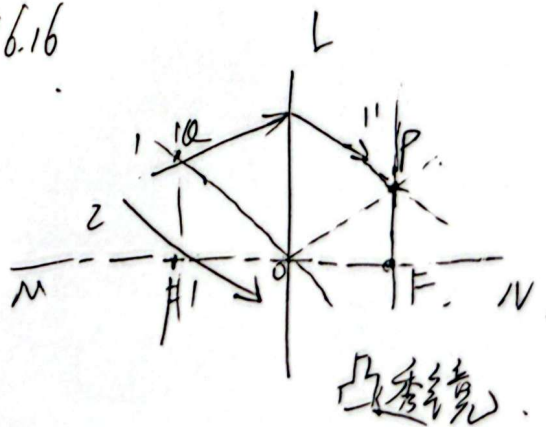
$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{10} + \frac{1}{v} = \frac{1}{-10}$$

$$v = -50\text{cm}$$



16.16



$$16.14. \frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{20} + \frac{1}{v} = \frac{1}{10}$$

$$v = 20\text{cm}$$

$$m = -\frac{v}{u} = -1$$

对于上半部分而言, 成像位于A的光轴上方1.00cm处.

对于下半部分B而言, 成像位于B的光轴下方1.00cm处

